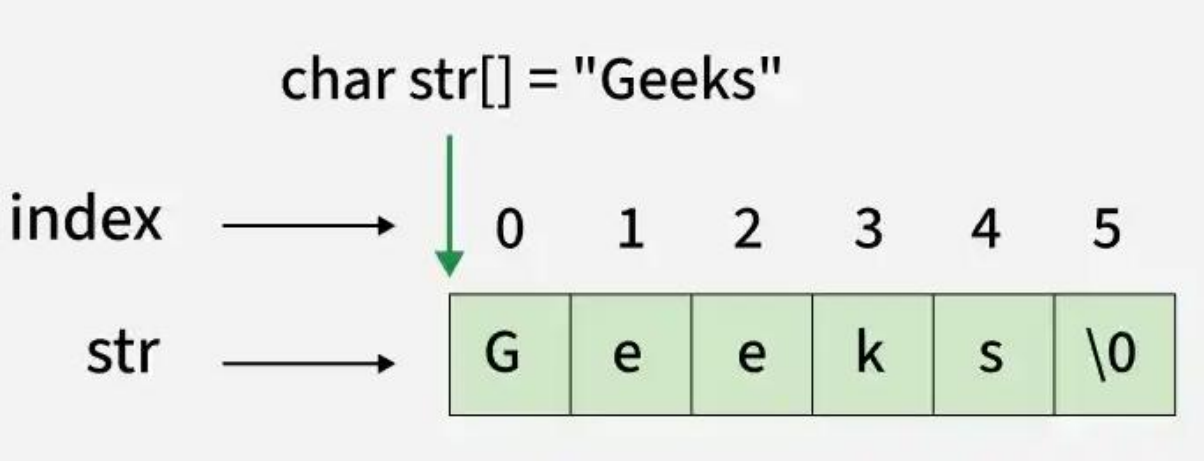
String Handling

String declaration

A string in C is a sequence of characters stored in a character array and terminated by the null character '\0'




```
#include <stdio.h>
int main() {
    char str[] = "Computer";
    printf("The string is: %s\n", str);
    return 0;
}
```
The string is: Computer

```
#include <stdio.h>
int main() {
    char str[] = {'R', 'a', 'v', 'i', '\0'};
    for(int i=0;i<5;i++)
    printf("%c", str[i]);
    return 0;
}
```
Ravi

```
#include <stdio.h>
int main() {
    char str[] = {'R', 'a', 'v', 'i', '\0'};
    printf("%s", str);
    return 0;
}
```
Ravi

```
#include <stdio.h>
int main() {
    char str[] = {'R', 'a', 'v', 'i'};
    printf("%s", str);
    return 0;
}
```
Ravi��k

```
#include <stdio.h>
int main() {
    char str[3]="ABC";
    printf("%s", str);
    return 0;
}
```
ABC???=

```
#include <stdio.h>
int main() {
    char str[3]="AB";
    printf("%s", str);
    return 0;
}
```
AB

```
#include <stdio.h>
int main() {
    char str[3]="ABC";
    for(int i=0;i<3;i++)
    printf("%c", str[i]);
    return 0;
}
```
ABC

```
#include <stdio.h>
int main() {
    char str[3]={'A','B','C'};
    for(int i=0;i<3;i++)
    printf("%c", str[i]);
    return 0;
}
```
ABC

Accessing Characters

```
#include <stdio.h>
int main() {
     char str[] = "Computer";
        printf("%c", str[0]);
    return 0;
}
```
C

Update

```
#include <stdio.h>
int main() {
    char str[] = "Mohan";
    str[0] = 'R';
    printf("%s", str);
    return 0;
}
```
Rohan

String Length

```
#include <stdio.h>
#include<string.h>
int main() {
    char str[] = "Software";

    printf("%d", strlen(str));
    return 0;
}
```

8

String Input and Output

```
#include<stdio.h>
int main() {
    char str[6];
    scanf("%s",str);
    printf("%s",str);
    return 0;
}
```

Undefined behaviour

```
#include<stdio.h>
int main() {
    char str[6];
    scanf("%5s",str);
    printf("%s",str);
    return 0;
}
```

Computer
Output
compu

RR BB
Output
RR

Using fgets()

```
#include <stdio.h>
int main() {
    char str[20];
    fgets(str, 20, stdin);
    printf("%s", str);
    return 0;
}
```

Ashoka Rajan
Ashoka Rajan

```
#include <stdio.h>
int main()
{
```

```
    char name[50];
    printf("Enter your full name: ");
    fgets(name, sizeof(name), stdin);
    printf("Your name is %s", name);
    return 0;
}
```

String Library Functions

Function	Purpose	Example
`strlen()`	Finds string length	`strlen("Hello")` → `5`
`strcpy()`	Copies one string to another	`strcpy(b, a)`
`strcat()`	Joins two strings	`strcat(a, b)`
`strcmp()`	Compares two strings	`strcmp(a, b)`

```
#include <stdio.h>
#include <string.h>
int main() {
    char a[30] = "Hello";
    char b[30] = "World";
    printf("Length = %d\n", strlen(a));
    strcpy(b, a);
    printf("After copying: %s\n", b);
    strcat(a, " Hai");
    printf("After joining: %s\n", a);
    printf("Comparison = %d\n", strcmp("c", "b"));
    printf("Comparison = %d\n", strcmp("Hello", "Welcome"));
    return 0;
}
Length = 5
After copying: Hello
After joining: Hello Hai
Comparison = 1
Comparison = -1
```